

(1001CJA102120063)

Test Pattern



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
04-04-2021

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE-II)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	C	B	A	D	A	A,C	A,B,C	B,C	A,C
	Q.	11	12								
	A.	A,C,D	B,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	4.80	1.25	0.02	8.48	24.00	0.50				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	D	C	D	D	A,C	B,D	A,C,D	A,C,D
	Q.	11	12								
	A.	A,B,C	A,B,C,D								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	126.45	5.90 to 5.92	17.4	234.25	10.00	9.00				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	D	C	B	C	C	A,D	A,B,C,D	B,C,D	A,B
	Q.	11	12								
	A.	B,C,D	A,B								
SECTION-II	Q.	1	2	3	4	5	6				
	A.	2.09	2.00	34.00	6.75	22.00	3.00				

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ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6				
	A.	A,B,D	A,C,D	B,D	A,D	A,C	A,D				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	15.00	8.00	0.00	7.66 to 7.67	35.00	8.00				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	5	2	8	4	8	5				

PART-2 : CHEMISTRY

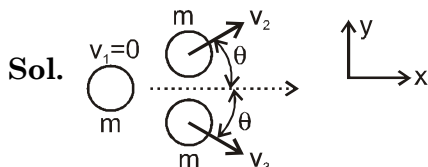
SECTION-I	Q.	1	2	3	4	5	6				
	A.	A,C,D	A,B,C	A,B,D	A,B,C,D	A,B,C,D	B,C				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	5.00	0.00	0.00	0.02	2.00	6.00				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	5	2	3	9	8	6				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6				
	A.	B,D	A,C	A,B,C	A,B,D	C,D	B,C,D				
SECTION-II	Q.	1	2	3	4	5	6				
	A.	0.45	45.95	31.25	1.25	1.00	1.75				
SECTION-III	Q.	1	2	3	4	5	6				
	A.	0	1	0	3	7	2				

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE-II)**PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (D)



After collision by momentum conservation

Along y-axis

$$0 = 0 + mv_2 \sin \theta - mv_3 \sin \theta$$

$$\Rightarrow v_2 = v_3$$

Along x-axis

$$mv = 0 + mv_2 \cos \theta + mv_3 \cos \theta$$

$$mv = 2m v_2 \cos \theta$$

$$v_2 = \frac{v}{2 \cos \theta}$$

$$\text{so } v_2 = v_3 > \frac{v}{2} \because \cos \theta < 1$$

2. Ans. (C)

Sol. Initially the potential at centre of sphere is

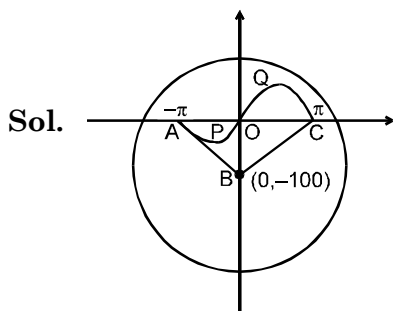
$$V_c = \frac{1}{4\pi\epsilon_0} \frac{Q}{x} + \frac{1}{4\pi\epsilon_0} \frac{2Q}{x} = \frac{1}{4\pi\epsilon_0} \frac{3Q}{x}$$

After the sphere is grounded, potential at centre becomes zero. Let the net charge on sphere finally be q .

$$\therefore \frac{1}{4\pi\epsilon_0} \frac{q}{r} + \frac{1}{4\pi\epsilon_0} \frac{3Q}{x} = 0 \text{ or } q = -\frac{3Q}{x} r$$

$$\therefore \text{The charge flowing out of sphere is } \frac{3Qr}{x}.$$

3. Ans. (B)



Connect centre B with the two ends points A & C

of the curves, by conducting rods.

 $\therefore$ Electric lines of force will be perpendicular to AB & CB. $\therefore$ E.M.F. developed in the loop, BAPOQCB

will be = E.M.F. developed in the curve APOQC.

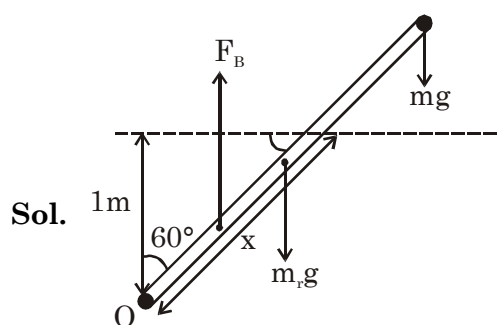
Now Flux in loop ABCQOPA = Φ [Area of ΔABC - Area of loop APOA + Area of loop OCOA] = Φ [Area of ΔABC]

$$= \frac{1}{2} \times (100) \times (2\pi)$$

$$= 100\pi = 314 \text{ m}^2 \therefore \Phi = 314 \text{ B}$$

$$\therefore \epsilon = 314 \frac{dB}{dt} = 314 \text{ V.}$$

4. Ans. (A)



$$\tau_{\text{net}_0} = 0$$

$$m_r g \times \frac{l}{2} \sin 60 + \frac{600}{1000} \times g \times l \sin 60 - F_B \times \frac{x}{2} = 0$$

$$F_B = \rho_L \times (A \times x) \times g$$

$$m_r = \rho_0 (A \times l)$$

By putting the values we get $A = 90 \text{ cm}^2$

5. Ans. (D)

Sol. For no ray to emerge out of side PR

$$A > 2C \Rightarrow \sin \frac{A}{2} > \sin C \Rightarrow \sin \frac{A}{2} > \frac{\sqrt{3}}{2}$$

or $A > 120^\circ$

6. Ans. (A)

Sol. At equilibrium $\frac{P_{\text{radiated}}}{A_{\text{sphere}}} = \frac{P_{\text{received}}}{A_{\text{disc}}}$

$$\frac{\sigma A_{\text{sun}} T_{\text{sun}}^4}{A_{\text{sphere}}} = \sigma T_{\text{disc}}^4$$

$$T_{\text{disc}} = T_{\text{sun}} \times \left(\frac{r_{\text{sun}}}{r_{\text{sun}} + d} \right)^{1/2}$$

$$T_{\text{disc}} = 6200 \times \left(\frac{6.8 \times 10^8}{1.5 \times 10^{11} + 6.8 \times 10^8} \right)^{1/2}$$

$$T_{\text{disc}} = 420.5 \text{ K}$$

7. Ans. (A,C)

Sol. K_α , K_β depends upon energy difference between specific orbit and not on applied potential difference.

8. Ans. (A,B,C)

Sol. Here $10 - T_2 = 10a$... (i)

$$T_2 - T_1 - 0.3 \times 2g = 3a$$
 ... (ii)

$$T_1 - 0.3 \times 2g = 2a$$

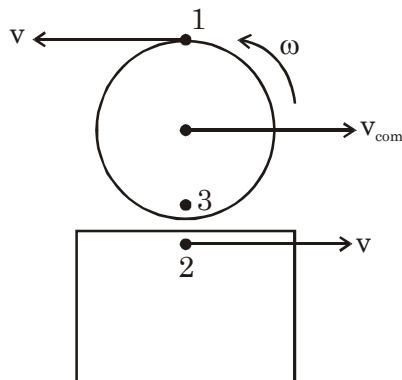
$$\text{Summing up } 10g - 0.3 \times 4 \times g = 15a$$

$$\text{i.e. } a = 5.86 \text{ ms}^{-2}$$

$$T_2 = 10 \times 9.8 - 10 \times 5.86 \text{ ms}^{-2} = 41.4 \text{ N}$$

$$T_1 = 2 \times 5.86 + 0.6 \times 9.8 = 17.7 \text{ N}$$

9. Ans. (B,C)



Sol.

$$v_1 = v_{\text{com}} - R\omega = -v$$

$$v_2 = v_3 = v_{\text{com}} + R\omega = +v$$

$$v_{\text{com}} = 0, \omega = v/R$$

10. Ans. (A, C)

Sol. Distance of mean position from water level
= immersed length
= maximum amplitude for equilibrium

$$\rho \times 60 \times a \times g = 3\rho L a g$$

maximum amplitude = L = immersed length = 20 cm

$$T = 2\pi \sqrt{\frac{m}{3\rho a g}}$$

11. Ans. (A,C,D)

Sol. $\frac{\lambda}{4} = 0.1 \Rightarrow \lambda = 0.4 \text{ m}$

from graph $\Rightarrow T = 0.2 \text{ sec.}$ and amplitude of standing wave is $2A = 4 \text{ cm.}$

Equation of the standing wave

$$y(x, t) = -2A \cos \left(\frac{2\pi x}{0.4} \right) \cdot \sin \left(\frac{2\pi t}{0.2} \right) \text{ cm}$$

$$y(x = 0.05, t = 0.05) = -2\sqrt{2} \text{ cm}$$

$$y(x = 0.04, t = 0.025) = -2\sqrt{2} \cos 36^\circ$$

$$\text{speed} = \frac{\lambda}{T} = 2 \text{ m/sec.}$$

$$V_y = \frac{dy}{dt} = -2A \times \frac{2\pi}{0.2} \cos \left(\frac{2\pi x}{0.4} \right) \cdot \cos \left(\frac{2\pi t}{0.2} \right)$$

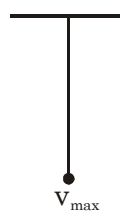
$$V_y = (x = \frac{1}{15} \text{ m}, t = 0.1) = 20\pi \text{ cm/sec}$$

12. Ans. (B,D)



Sol.

$$a_\tau = \frac{F_\tau}{m} = g, a_c = \frac{v^2}{R} = 0$$



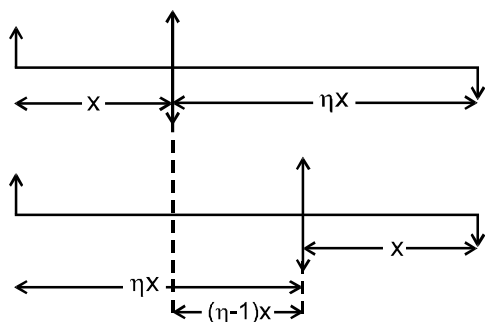
$$a_{c_{\text{max}}} = \frac{v_{\text{max}}^2}{R} \text{ at lowest point}$$

$$a_t = \frac{F_\tau}{m} = 0$$

SECTION-II

1. Ans. 4.80

Sol.



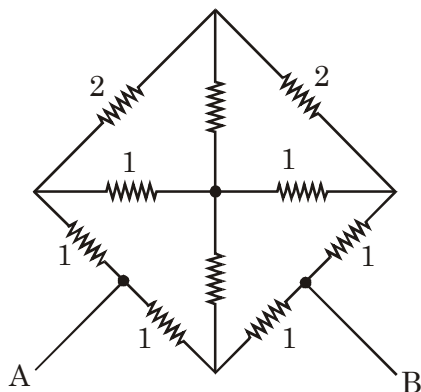
using lens formula :

$$\frac{1}{\eta x} + \frac{1}{x} = \frac{1}{f} \Rightarrow f = \frac{\eta x}{\eta + 1}$$

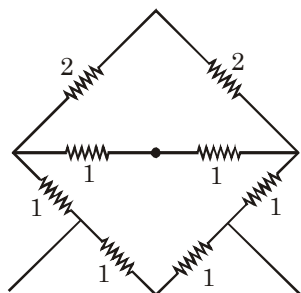
so the given ratio is $\frac{(\eta^2 - 1)}{\eta}$

2. Ans. 1.25

Sol.



Since it is symmetric about AB



$$R_{AB} = 1.25$$

3. Ans. 0.02

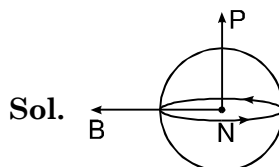
Sol. Energy stored in capacitor when it is charged upto $2V = \frac{1}{2} 10 \cdot 2^2 = 20\mu J = u_1$ (suppose)
Energy stored in capacitor when it is charged upto $4V = \frac{1}{2} 10 \cdot 4^2 = 80\mu J = u_2$ (suppose)

Increase in charge = $40 - 20 = 20\mu C$

Energy drawn from cell = $20 \times 4 = 80\mu J = u$ (suppose)

$$\begin{aligned} \text{Heat produced} &= u_1 + u - u_2 \\ &= 20 + 80 - 80 \\ &= 20 \mu J \end{aligned}$$

4. Ans. 8.48



Sol.

Torque on the (coil + sphere) due to flow of charge through coil is

$$= |\vec{p} \times \vec{B}| \text{ (where } \vec{p} \text{ is the dipole moment}$$

of the coil and $\vec{B}$ is the geomagnetic field)

$$= i N \pi r^2 B = I \frac{d\omega}{dt}$$

$$\therefore d\omega = \frac{N \pi r^2 B}{I} i dt$$

$$\text{or } \omega = \frac{N \pi r^2 B}{\frac{2}{3} m r^2} \int_0^{\Delta t} i dt = \frac{3 N \pi B Q}{2} \text{ Ans.}$$

$$\text{Ans: } \omega = \frac{3 B N \pi Q}{2 M} = 2.7 \pi \times 10^{-2} \text{ rad/s.}$$

5. Ans. 24.00

Sol. Shift of fringe pattern = $(\mu - 1) \frac{tD}{d}$

$$\therefore \frac{30 D (4800 \times 10^{-10})}{d} = (0.6) t \frac{D}{d}$$

$$30 \times 4800 \times 10^{-10} = 0.6$$

$$t = \frac{30 \times 4800 \times 10^{-10}}{0.6}$$

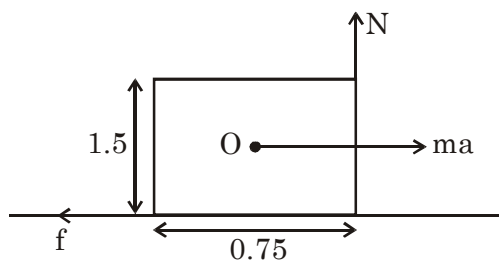
$$= \frac{1.44 \times 10^{-5}}{0.6} = 24 \times 10^{-6}$$

6. Ans. 0.50



Sol.

w.r.t truck F.B.D.



$$\tau_0 = 0$$

$$f \times \frac{1.5}{2} = N \times \frac{0.75}{2}$$

$$f = ma, N = mg$$

$$a = 10 \times \frac{0.75}{1.5} = 5 \text{ m/s}^2$$

$$f = \mu N$$

$$\mu_{\min} = \frac{ma}{mg} = \frac{5}{10} = 0.5$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(A)

$$\begin{aligned} \text{Sol. BE of reactant} &= 6 \times E_{\text{C-H}} + 2E_{\text{C-C}} + E_{\text{C}\equiv\text{C}} \\ &= 6 \times E_{\text{C-H}} + 2 \times 348 + 835 \\ &= 6 \times E_{\text{C-H}} + 1531 \end{aligned}$$

$$\begin{aligned} \text{BE of Product} &= 6E_{\text{C-H}} + 3E_{\text{C-C}} + E_{\text{C}\equiv\text{C}} \\ &= 6E_{\text{C-H}} + 3 \times 348 + 610 \\ &= 6E_{\text{C-H}} + 1654 \end{aligned}$$

$$\Delta H = 1531 - 1654 = -123 \text{ kJ}$$

2. Ans.(D)

- Sol. (i) In a period inert gas has highest 1st ionization energy
 (ii) EGE of alkali has negative value
 (iii) Metal oxide in higher oxidation state are generally acidic in nature
 (iv) In sp^3d^2 hybridization $d_{x^2-y^2}$ and d_{z^2} orbitals are involved

3. Ans.(D)

4. Ans.(C)

$$\text{Sol. } \frac{-dp}{dt} = K P_{\text{NO}}^a P_{\text{H}_2}^b$$

For 1st experiment, hydrogen pressure is

$$\text{constant } \frac{-dp}{dt} = K^1 P_{\text{NO}}^a$$

$$2.25 \text{ torr/sec} = k^1 \alpha (450 \text{ torr})^a \text{ --- (I)}$$

$$2.25 \text{ torr/sec} = k^1 \alpha (450 \text{ torr})^a \text{ --- (II)}$$

$$\text{On dividing } \frac{2.25}{0.25} = \left(\frac{450}{150} \right)^a = 3^a$$

$$9 = 3^a \Rightarrow a = 2$$

For IInd experiment, NO pressure is constant

$$1.90 = k^{ii} (291)^b \text{ --- (III)}$$

$$0.951 = k^{ii} (145)^b \text{ --- (IV)}$$

$$\text{On Dividing } \frac{1.90}{0.95} = \left(\frac{291}{145} \right)^b$$

$$2 = 2^b \Rightarrow b = 1$$

$$\boxed{\frac{-dp}{dt} = k P_{\text{NO}}^2 P_{\text{H}_2}}$$

5. Ans.(D)

Sol. Dipole moment $\mu = e \times d$ and for polyatomic molecules, vector addition is applied.

6. Ans.(D)

Sol. Reaction proceed via SN_1 and E_1 mechanism as carbocation will be the intermediate and so all the products will be feasible.

7. Ans.(A, C)

Sol. In lyophilic colloids dispersion phase shows high affinity with dispersion medium and molecules of DP are large enough to be close to lower limit of colloidal range thus they pass into colloidal state.

8. Ans. (B,D)

9. Ans. (A,C,D)

10. Ans. (A,C,D)

11. Ans. (A,B,C)

12. Ans. (A,B,C,D)

Sol. (i) Reaction used to protect certain carbonyl groups

- (ii) Reaction is not favourable with monohydric alcohol because ENTROPY decreases.
- (iii) In acidic medium acetal is not stable.

SECTION-II

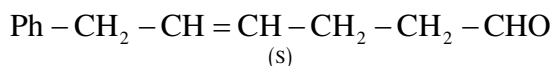
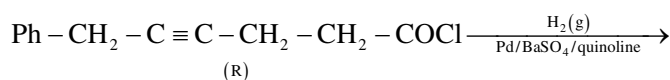
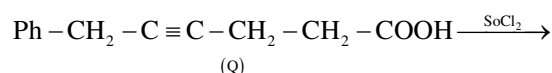
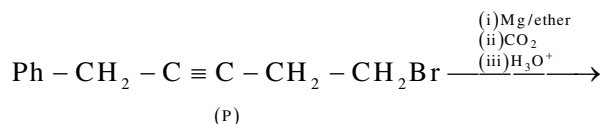
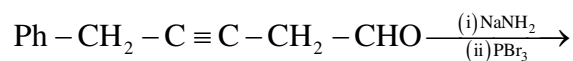
1. **Ans.(126.45)**

2. **Ans.5.90 to 5.92**

Sol. In $[\text{Mn}(\text{SCN})_6]^{4-}$ Mn^{+2} has d^5 configuration

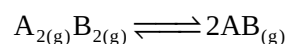
$$\begin{aligned}\text{So that magnetic moment} &= \sqrt{5 \times 7} \\ &= \sqrt{35} \\ &= 5.916\end{aligned}$$

3. **Ans.(17.4)**



$$\begin{aligned}\text{M wt.} &= 72 + 5 + 14 + 26 + 28 + 29 = 174 \\ \text{M}/10 &= 17.4\end{aligned}$$

4. **Ans.(234.25)**



$$\Delta S^\circ_{\text{reaction}} = [2 \times \Delta_f S^\circ_{\text{AB}}] - [\Delta_f S^\circ_{\text{A}_{2(\text{g})}} + \Delta_f S^\circ_{\text{B}_{2(\text{g})}}]$$

$$\Delta S^\circ = 25 \text{ JK}^{-1} \text{ mole}^{-1}$$

$$\Delta H^\circ_{\text{reaction}} = [2\Delta_f H^\circ_{\text{AB}}] - [\Delta_f H^\circ_{\text{A}_2} + \Delta_f H^\circ_{\text{B}_2}]$$

$$\Delta H^\circ = 270.5 \text{ kJ mol}^{-1}$$

$$Q_{\text{eq}} = \frac{P_{\text{AB}(\text{g})}^2}{P_{\text{A}_2} \times P_{\text{B}_2}}$$

$$Q_{\text{eq}} = \frac{(10^{-4})^2}{10^{-2} \times 10^{-1}} = 10^{-5}$$

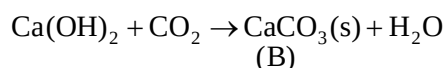
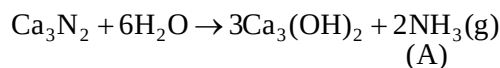
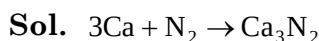
$$\Delta G = \Delta G^\circ + RT \ln Q$$

$$\Delta G = \Delta H^\circ - T\Delta S^\circ + RT \ln Q$$

$$\Delta G = 270.5 \text{ kJ} - 7.5 \text{ kJ} + 5.75 \times (-5)$$

$$\Delta G = 234.25 \text{ kJ/mole}$$

5. **Ans.(10.00)**

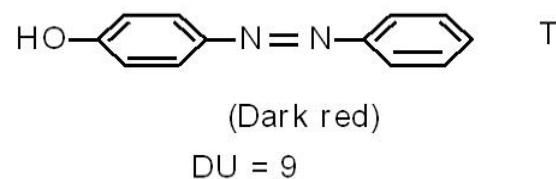
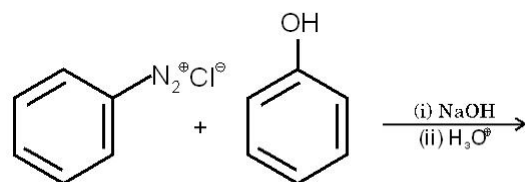
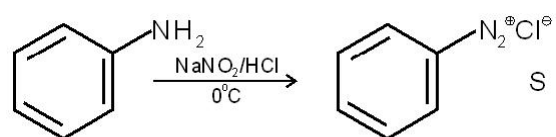
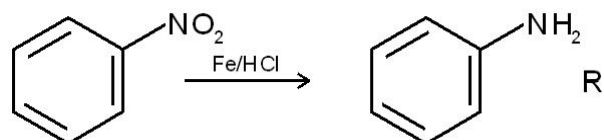
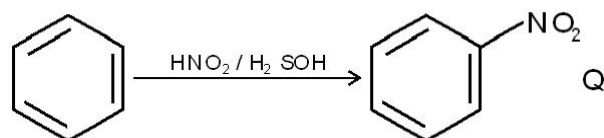
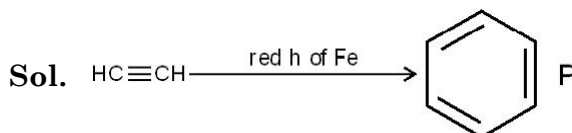


$$4\text{gm Ca} \equiv 0.1\text{mole Ca}$$

$$\equiv 0.1\text{mole CaCO}_3$$

$$\equiv 10\text{gm CaCO}_3$$

6. **Ans.(9.00)**



PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (C)**

Sol. $2PA = 3PB$

$$\Rightarrow 2\sqrt{(x-9\sqrt{2})^2 + (y-9\sqrt{2})^2} = 3\sqrt{(x-4\sqrt{2})^2 + (y+4\sqrt{2})^2}$$

$$\Rightarrow 4[x^2 + y^2 - 18\sqrt{2}x - 18\sqrt{2}y + 324] = 9[x^2 + y^2 - 8\sqrt{2}x - 8\sqrt{2}y + 64]$$

$$\Rightarrow 5(x^2 + y^2) = 720$$

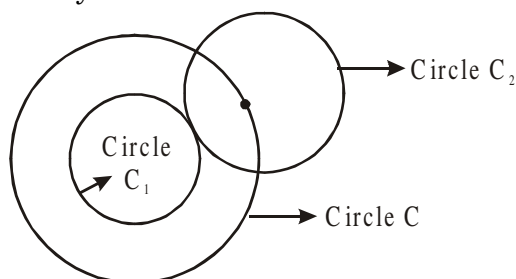
$$\Rightarrow x^2 + y^2 = 144$$

Let this circle be C_1 , then circle C_2 of radius 3 touches C_1 externally.

$\therefore$ locus of centre of circle C_2 will also be a circle 'C' concentric with circle C_1 and of radius = radius of circle $C_1 + 3 = 12 + 3$

$\therefore$ locus of centre of circle C_2

i.e. equation of circle 'C' is $x^2 + y^2 = 15^2$



2. **Ans. (D)**

Sol. $2021 = 43 \times 47 = abcd$

$\therefore$ No. of ways of distributing an object (43 or 47) to 4 distinct places {a,b,c and d} is 4

$\therefore$ No. of positive integral solutions of $abcd = 43 \times 47$ is $4 \times 4 = 16$

For number of integral solutions

Among a,b,c,d either none or any 2 or all 4 can be negative

$\therefore$ The number of integral solutions of $abcd = 2021 = 43 \times 47$ is

$$({}^4C_0 + {}^4C_2 + {}^4C_4) \times 16 = 8 \times 16 = 128$$

3. **Ans. (C)**

Sol. Since, only (C) satisfies given definition $f\{f^{-1}(B)\} = B$ only, if $B \subseteq f(x)$

4. **Ans. (B)**

$$\text{Sol. } \frac{x^2 - x + 2}{(x+1)(x-1)^3} = \frac{(t+1)^2 - (t+1) + 2}{(t+1+1)t^3}$$

{Substituting $x-1=t$ }

$$= \frac{2+t+t^2}{(2+t)t^3}$$

$$\begin{array}{r} 1 + \frac{t^2}{2} \\ 2+t \overline{) 2+t+t^2} \\ \underline{2+t} \\ x \\ t^2 + \frac{t^3}{2} \\ \underline{ + \frac{t^3}{2}} \\ x - \frac{t^3}{2} \end{array}$$

$$= \frac{(2+t)\left(1+\frac{t^2}{2}\right) - \frac{t^3}{2}}{(2+t)t^3} = \frac{1}{t^3} + \frac{1}{2t} - \frac{1}{2(2+t)} = \frac{1}{(x-1)^3} + \frac{1}{2(x-1)} - \frac{1}{2(x+1)}$$

$$\therefore \int \frac{x^2 - x + 2}{(x+1)(x-1)^3} dx = \int \frac{-1}{2(x+1)} + \frac{1}{2(x-1)} + \frac{1}{(x-1)^3} dx$$

$$= -\frac{1}{2} \ln|x+1| + \frac{1}{2} \ln|x-1| + \left(-\frac{1}{2}\right) \frac{1}{(x-1)^2} + E$$

$$\therefore A = -\frac{1}{2}, B = \frac{1}{2}, C = 0 \text{ and } D = -\frac{1}{2}$$

$$\therefore A + D = -1 = C - 2B$$

5. **Ans. (C)**

$$\text{Sol. } \lim_{x \rightarrow 0} \left(\frac{f(1+x)}{f(1)} \right)^{\frac{1}{x}} = e^{\lim_{x \rightarrow 0} \left\{ \frac{f(1+x)}{f(1)} - 1 \right\} \cdot \frac{1}{x}} = e^{\lim_{x \rightarrow 0} \frac{f(1+x) - f(1)}{x} \cdot \frac{1}{f(1)}}$$

Applying L Hospital's rule

$$= e^{\lim_{x \rightarrow 0} \frac{f'(1)}{f(1)}} = e^{\frac{2\sqrt{2}}{\sqrt{2}}} = e^2$$

6. **Ans. (C)**

$$\text{Sol. } \frac{2 \sin \theta}{\cos 3\theta} = \frac{2 \sin \theta \cos \theta}{\cos \theta \cos 3\theta} = \frac{\sin(3\theta - \theta)}{\cos \theta \cos 3\theta} = \tan 3\theta - \tan \theta$$

similarly

$$\frac{2 \sin 3\theta}{\cos 9\theta} = \tan 9\theta - \tan 3\theta$$

$$\frac{2 \sin 9\theta}{\cos 27\theta} = \tan 27\theta - \tan 9\theta$$

$$\therefore f(\theta) = (\tan 3\theta - \tan \theta) + (\tan 9\theta - \tan 3\theta) + (\tan 27\theta - \tan 9\theta) + \tan \theta$$

$$\Rightarrow f(\theta) = \tan 27\theta$$

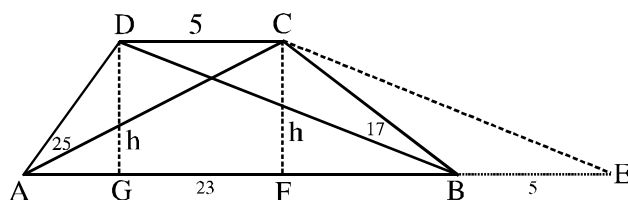
$$\Rightarrow \theta \in \left[0, \frac{\pi}{108}\right]$$

$$\Rightarrow 27\theta \in \left[0, \frac{\pi}{4}\right]$$

$$\therefore f(\theta) = \tan 27\theta \in [0, 1]$$

7. **Ans. (A,D)**

Sol.



In $\triangle ACE$

$$a = AE = 23 + 5 = 28$$

$$b = AC = 25$$

$$c = CE = 17$$

$$\therefore s = \frac{1}{2}(a + b + c) = \frac{1}{2}(28 + 25 + 17) = 35$$

$$\Delta = \frac{1}{2}AE \cdot h = \sqrt{35(35-28)(35-25)(35-17)}$$

$$\Rightarrow \frac{1}{2} \cdot 28 \cdot h = \sqrt{5 \cdot 7 \cdot 7 \cdot 5 \cdot 2 \cdot 2 \cdot 3 \cdot 3}$$

$$\Rightarrow 14h = 5 \cdot 7 \cdot 2 \cdot 3 \quad \therefore h = 15$$

$$\therefore \text{In } \triangle CEF: EF = \sqrt{17^2 - 15^2} = 8$$

$$\therefore BF = 8 - 5 = 3$$

$$\therefore |BC| = \sqrt{h^2 + 3^2}$$

$$= \sqrt{225 + 9} = \sqrt{234}$$

$$= 3\sqrt{26}$$

From $\triangle ACF$

$$AF = \sqrt{25^2 - h^2}$$

$$= \sqrt{25^2 - 15^2} = 20$$

$$\therefore AG = AF - GF = 20 - 5 = 15$$

$$\therefore AD = \sqrt{h^2 + AG^2} = \sqrt{15^2 + 15^2}$$

$$= 15\sqrt{2}$$

8. Ans. (A,B,C,D)

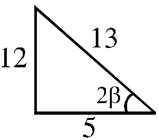
Sol. Let $\alpha = \frac{1}{2}\cos^{-1}\frac{3}{5}$ and $\beta = \frac{1}{2}\tan^{-1}\frac{12}{5}$

$$\text{then } \cos 2\alpha = \frac{3}{5} \text{ \& } \tan 2\beta = \frac{12}{5}$$

$$\left\{ 0 < \alpha, \beta < \frac{\pi}{2} \right\}$$

$$\therefore \tan \alpha = \sqrt{\frac{1 - \cos 2\alpha}{1 + \cos 2\alpha}} = \sqrt{\frac{1 - \frac{3}{5}}{1 + \frac{3}{5}}} = \sqrt{\frac{1}{4}} = \frac{1}{2}$$

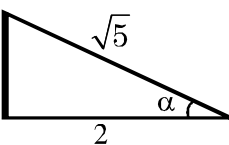
$$\therefore \alpha = \tan^{-1}\left(\frac{1}{2}\right)$$

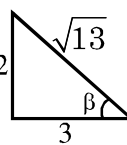
Also  $\Rightarrow \cos 2\beta = \frac{5}{13}$

$$\therefore \tan \beta = \sqrt{\frac{1 - \cos 2\beta}{1 + \cos 2\beta}} = \sqrt{\frac{1 - \frac{5}{13}}{1 + \frac{5}{13}}} = \sqrt{\frac{8}{18}} = \frac{2}{3}$$

$$\therefore \beta = \tan^{-1}\left(\frac{2}{3}\right)$$

$$\therefore \alpha + \beta = \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{2}{3}\right)$$

Also 



$$\Rightarrow \alpha = \sin^{-1} \frac{1}{\sqrt{5}} = \cos^{-1} \frac{2}{\sqrt{5}} \text{ and } \beta = \sin^{-1} \frac{2}{\sqrt{13}} = \cos^{-1} \frac{3}{\sqrt{13}}$$

$$\therefore \alpha + \beta = \sin^{-1} \frac{1}{\sqrt{5}} + \cos^{-1} \frac{3}{\sqrt{13}}$$

$$= \sin^{-1} \frac{2}{\sqrt{13}} + \cos^{-1} \frac{2}{\sqrt{5}}$$

$$\text{Also } \tan^{-1} \frac{1}{4} + \tan^{-1} \frac{2}{9} + \cot^{-1} \left(\frac{3}{2}\right) = \tan^{-1} \left(\frac{\frac{1}{4} + \frac{2}{9}}{1 - \frac{1}{4} \cdot \frac{2}{9}} \right) + \tan^{-1} \frac{2}{3}$$

$$= \tan^{-1} \left(\frac{17/36}{34/36} \right) + \tan^{-1} \left(\frac{2}{3} \right)$$

$$= \tan^{-1} \left(\frac{1}{2} \right) + \tan^{-1} \left(\frac{2}{3} \right) = \alpha + \beta$$

9. Ans. (B,C,D)

Sol. Given parabola $y = x^2 + ax + 1$

$$\Rightarrow y = \left(x + \frac{a}{2}\right)^2 + 1 - \frac{a^2}{4}$$

As no any point of the parabola lies below x axis, $1 - \frac{a^2}{4} \geq 0$

$$\therefore \frac{a^2}{4} \leq 1$$

$$\therefore a \in [-2, 2]$$

$$\therefore \text{No. of integral values of } a = 5$$

The tangent to $y = x^2 + ax + 1$ at $(0, 1)$ is $T = 0$

$$\Rightarrow \frac{y+1}{2} = x \cdot 0 + a \left(\frac{x+0}{2} \right) + 1$$

$$\Rightarrow y+1 = ax+2 \Rightarrow y = ax+1$$

This is tangent to $x^2 + y^2 = r^2$ also

$$\therefore 1^2 = r^2(a^2 + 1) \Rightarrow r^2 = \frac{1}{a^2 + 1}$$

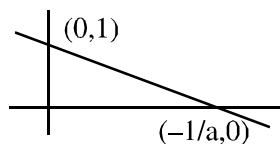
$$\therefore \text{for maximum value of } a, r^2 = \frac{1}{4+1}$$

$$\therefore r = \frac{1}{\sqrt{5}}$$

for maximum value of radius r , a^2 must be minimum i.e. $a^2 = 0$, for this

$a = \text{slope of tangent} = 0$

$$\text{Also, area enclosed by the tangent } y = ax + 1 \text{ with coordinate axes} = \frac{1}{2} \cdot 1 \cdot \left| -\frac{1}{a} \right| = \frac{1}{2|a|}$$

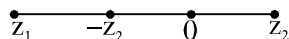


$$\therefore \text{Minimum area (when } |a| = 2)$$

$$= \frac{1}{2 \times 2} = \frac{1}{4}$$

10. Ans. (A,B)

$$\text{Sol. (1) } |z_1| + |z_2| = 4 \Rightarrow |z_1| + |-z_2| = |z_1 + (-z_2)|$$

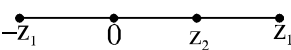


$\therefore z_1$ & $(-z_2)$ are collinear with origin such that z_1 & z_2 both lie towards the same side of origin.

$\therefore z_1, z_2$ and origin will also be collinear such that z_1 and z_2 will lie opposite to the side of

$$\text{origin } \therefore \arg\left(\frac{z_1}{z_2}\right) = \pi$$

$$(2) |z_1| + |z_2| = |z_1 + z_2|$$

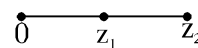


$\therefore z_1$ & z_2 are collinear with origin such that z_1 & z_2 both lie towards the same side of origin

$$\therefore \arg\left(\frac{z_1}{z_2}\right) = 0$$

$$(3) |z_1| - |z_2| = -4 \Rightarrow |z_1| - |z_2| = -|z_1 - z_2|$$

$$\therefore |z_1| + |z_2 - z_1| = |z_2|$$

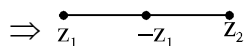


$\therefore z_1, z_2$ and origin will

such that z_1 and z_2 will be towards the same side of origin $\therefore \arg\left(\frac{z_1}{z_2}\right) = 0$

$$(4) |z_2| - |z_1| = |z_1 + z_2|$$

$$\Rightarrow |z_1 + z_2| + |z_1| = |z_2|$$



$\therefore z_1, z_2$ and origin will be collinear such that z_1 and z_2 will lie opposite to the side of origin $\therefore$

$$\arg\left(\frac{z_1}{z_2}\right) = \pi$$

11. Ans. (B,C,D)

Sol. Let E = Easy, D = Difficult, T = True / false type & M = MCQ

Total no. of questions = 300 + 200 + 500 + 400 = 1400

$$P(E) = \frac{300+500}{1400} = \frac{8}{14} = \frac{4}{7}$$

$$P(T) = \frac{300+200}{1400} = \frac{5}{14}$$

$$P(D) = \frac{200+400}{1400} = \frac{6}{14} = \frac{3}{7}$$

$$P(M) = \frac{500+400}{1400} = \frac{9}{14}$$

$$P(E \cap T) = \frac{300}{1400} = \frac{3}{14}, P(D \cap T) = \frac{200}{1400} = \frac{1}{7}$$

$$P(E \cap M) = \frac{500}{1400} = \frac{5}{14}, P(D \cap M) = \frac{400}{1400} = \frac{2}{7}$$

$$\therefore P(E/M) = \frac{P(E \cap M)}{P(M)} = \frac{5/14}{9/14} = \frac{5}{9}$$

$$P(M/E) = \frac{P(E \cap M)}{P(E)} = \frac{5/14}{4/7} = \frac{5}{8}$$

$$P(D/T) = P\left(\frac{D \cap T}{P(T)}\right) = \frac{1/7}{5/14} = \frac{2}{5}$$

$$P(E \cap T) = \frac{300}{1400} = \frac{3}{14}$$

12. Ans. (A,B)

Sol. (1) $\text{adj}(\text{adj } A) = |A|^{n-2} A$ (common property)

$$(2) \text{adj}(AB)^T = \text{adj}(B^T A^T) = \text{adj } A^T \cdot \text{adj } B^T = (\text{adj } A)^T \cdot (\text{adj } B)^T$$

$$(3) |\text{adj}(\text{adj } A)| = \left| |A|^{n-2} A \right|$$

$$= \left(|A|^{n-2} \right)^n |A| = |A|^{(n-1)^2}$$

$$(4) \text{Let } C = A^2 B^2 - B^2 A^2$$

$$\text{Then } C^T = (A^2 B^2 - B^2 A^2)^T = (A^2 B^2)^T - (B^2 A^2)^T$$

$$= (B^2)^T \cdot (A^2)^T - (A^2)^T (B^2)^T$$

$$= (B^T)^2 (A^T)^2 - (A^T)^2 (B^T)^2$$

$$= B^2 (-A)^2 - (-A)^2 (B)^2$$

$$= B^2 A^2 - A^2 B^2$$

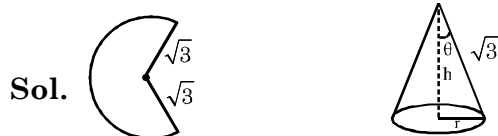
$$= -C$$

$\therefore C$ is skew symmetric matrix of odd order 'n'

$\therefore C$ is non-invertible.

SECTION-II

1. Ans. 2.09



$$r = \sqrt{3} \sin \theta$$

$$h = \sqrt{3} \cos \theta$$

$$v = \frac{1}{3} \pi r^2 h = \frac{\pi}{3} (\sqrt{3} \sin \theta)^2 \cdot \sqrt{3} \cos \theta$$

$$v = \pi \sqrt{3} \sin^2 \theta \cos \theta \quad \dots(1)$$

Applying AM – GM inequality in $\frac{\sin^2 \theta}{2}, \frac{\sin^2 \theta}{2}$ & $\cos^2 \theta$, we have

$$\frac{2 \left(\frac{\sin^2 \theta}{2} \right) + \cos^2 \theta}{3} \geq \sqrt[3]{\left(\frac{\sin^2 \theta}{2} \right)^2 \cos^2 \theta}$$

$$\therefore \sin^2 \theta \cdot \cos \theta \leq \sqrt{\frac{4}{27}} = \frac{2}{3\sqrt{3}}$$

$$\therefore V_{\max} = \pi \sqrt{3} \sin^2 \theta \cos \theta \big|_{\max} = \pi \sqrt{3} \cdot \frac{2}{3\sqrt{3}}$$

$$= \frac{2\pi}{3} \approx 2.09$$

2. Ans. 2.00

Sol. $\frac{x \ln x + \ln x}{x-1} > k \quad \forall x > 1$

Let $f(x) = \left(\frac{x+1}{(x-1)} \ln x \right)$

Then $f'(x) = \left(\frac{x+1}{(x-1)} \right) \cdot \frac{1}{x} + \ln x \cdot \frac{(x-1) \cdot 1 - (x+1) \cdot 1}{(x-1)^2}$

$$= \frac{(x+1)}{x(x-1)} - \frac{2 \ln x}{(x-1)^2}$$

$$= \frac{x^2 - 1 - 2x \ln x}{x(x-1)^2} = \frac{g(x)}{x(x-1)^2}$$

$$\therefore g(x) = x^2 - 1 - 2x \ln x$$

$$\therefore g'(x) = 2x - 0 - 2x \cdot \frac{1}{x} - 2 \cdot \ln x \cdot 1 = 2[(x-1) - \ln x]$$

$$\therefore g''(x) = 2 \left[1 - \frac{1}{x} \right] = 2 \left[\frac{x-1}{x} \right] > 0 \quad \forall x > 1$$

$$\therefore g'(x) \text{ is strictly increasing } \forall x > 1$$

$$\therefore g'(x) > g'(1) \quad \forall x > 1 \Rightarrow g'(x) > 0$$

$$\therefore g(x) \text{ is strictly increasing } \forall x > 1$$

$$\therefore g(x) > g(1) \quad \forall x > 1$$

$$\therefore g(x) > 0 \quad \forall x > 1$$

$$\therefore f'(x) = \frac{g(x)}{x(x-1)^2} > 0 \quad \forall x > 1$$

$$\therefore f(x) \text{ is strictly increasing } \forall x > 1$$

$$\therefore f(x) > f(1) \quad \forall x > 1$$

$$\therefore \text{Minimum value of } f(x)$$

$$= \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} \frac{(x+1) \ln x}{(x-1)}$$

$$= \lim_{y \rightarrow 0^+} (2+y) \frac{\ln(1+y)}{y} = 2$$

$$\therefore \text{Minimum value of } f(x) = 2$$

$$\therefore f(x) > 2 \quad \forall x > 1$$

$$\therefore K_{\max} = 2$$

3. Ans. 34.00

$$\text{Sol. } f'(x) = \frac{2x+1}{(x-1)^2+1} > 0 \quad \forall x \in \left(-\frac{1}{2}, 1\right] \text{ \& } f'(x) < 0 \quad \forall x \in \left[-1, -\frac{1}{2}\right)$$

$$\therefore f(x) \text{ strictly increases in } \left(-\frac{1}{2}, 1\right] \text{ \& } f(x) \text{ strictly decreases in } \left[-1, -\frac{1}{2}\right)$$

$$\therefore f(x)_{\min} = f\left(-\frac{1}{2}\right) \text{ and } f(x) \text{ will be maximum either at } x=1 \text{ or at } x=-1$$

$$\therefore f(x) = \int_0^x \frac{2(t-1)+3}{t^2-2t+2} dt$$

$$= \left[\ln(t^2-2t+2) + 3 \tan^{-1}(t-1) \right]_0^x$$

$$= \ln(x^2-2x+2) + 3 \tan^{-1}(x-1) - \ln 2 + \frac{3\pi}{4}$$

$$\therefore f\left(-\frac{1}{2}\right) = \ln\left(\frac{13}{4}\right) + 3 \tan^{-1}\left(-\frac{3}{2}\right) - \ln 2 + \frac{3\pi}{4}$$

$$= \ln\left(\frac{13}{8}\right) + 3\left(\tan^{-1} 1 - \tan^{-1} \frac{3}{2}\right)$$

$$= \ln\left(\frac{13}{8}\right) - 3 \tan^{-1}\left(\frac{1}{5}\right)$$

$$\therefore f(1) = 0 + 0 + \frac{3\pi}{4} - \ln 2 \text{ \& } f(-1) = \ln 5 - 3 \tan^{-1} 2 - \ln 2 + \frac{3\pi}{4}$$

$$= \left(\frac{3\pi}{4} - \ln 2\right) - (3 \tan^{-1} 2 - \ln 5)$$

$$= f(1) - (\text{a + ve quantity})$$

$$\therefore f(-1) < f(1)$$

$$\therefore f(x) \text{ maxima} = f(1) = \frac{3\pi}{4} - \ln 2$$

$$\therefore \text{Range of } f(x) = \left[\ln\left(\frac{13}{8}\right) - 3\tan^{-1}\frac{1}{5}, \frac{3\pi}{4} - \ln 2 \right]$$

$$a = \ln\frac{13}{8} - 3\tan^{-1}\frac{1}{5} \text{ \& } b = \frac{3\pi}{4} - \ln 2$$

$$\Rightarrow a + b = \ln\frac{13}{8} - \ln 2 + 3\left(\tan^{-1}1 - \tan^{-1}\frac{1}{5}\right)$$

$$= \ln\frac{13}{16} + 3\tan^{-1}\left(\frac{1 - \frac{1}{5}}{1 + \frac{1}{5}}\right)$$

$$= \ln\frac{13}{16} + 3\tan^{-1}\left(\frac{2}{3}\right)$$

$$= \ln\left(\frac{m}{n}\right) + 3\tan^{-1}\left(\frac{p}{q}\right)$$

$$\therefore m = 13 \quad n = 16$$

$$P = 2 \quad q = 3$$

$$\therefore m + n + p + q = 34$$

4. **Ans. 6.75**

Sol. $(x^2 - 11)(y + 1) = -4 \Rightarrow (x^2 - 11) \cdot (y + 1) = (-2) \cdot 2$

$$\therefore x^2 - 11 = -2, y + 1 = 1 \Rightarrow P(x_0, y_0) \equiv (3, 1)$$

It can easily be proved that no any other such point P may exist.

$$\therefore y = \frac{-4}{x^2 - 11} - 1 \quad \therefore \frac{dy}{dx} = \frac{4 \cdot 2x}{(x^2 - 11)^2} - 0$$

$$\text{At } P(x_0, y_0) \equiv P(3, 1)$$

$$\text{The slope of tangent, } m = \frac{4 \cdot 2 \cdot 3}{(3^2 - 11)^2} = 6$$

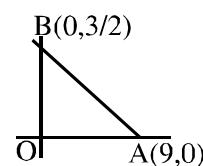
$$\therefore \text{the slope of normal} = -\frac{1}{m} = -\frac{1}{6}$$

$$\therefore \text{Equation of normal at } P(3, 1)$$

$$y - 1 = -\frac{1}{6}(x - 3) \Rightarrow 6y - 6 + x - 3 = 0$$

$$\Rightarrow x + 6y = 9$$

$$\Rightarrow \frac{x}{9} + \frac{y}{3/2} = 1$$



$$\therefore \text{Area of } \triangle OAB = \frac{1}{2} \cdot 9 \cdot \frac{3}{2} = \frac{27}{4} = 6.75$$

$$\text{or Area of } \triangle OAB = \int_0^9 y \, dx = \int_0^9 \left(\frac{9-x}{6} \right) dx$$

$$= \frac{1}{6} \left[\frac{(9-x)^2}{-2} \right]_0^9 = \frac{1}{6} \left(0 - \frac{-81}{2} \right)$$

$$= \frac{81}{12} = \frac{27}{4} = 6.75$$

5. Ans. 22.00

Sol. $\frac{dy}{dx} + \frac{6y}{x} = 3x^2y^2$

$$\Rightarrow \frac{-1}{y^2} \frac{dy}{dx} + \left(\frac{-6}{x} \right) \cdot \frac{1}{y} = -3x^2$$

$$\frac{1}{y} = t \quad \therefore \frac{-1}{y^2} \frac{dy}{dx} = \frac{dt}{dx}$$

$$\Rightarrow \frac{dt}{dx} - \frac{6}{x} \cdot t = -3x^2$$

$$\text{I.F.} = e^{\int \frac{-6}{x} dx} = \frac{1}{x^6}$$

$\therefore$ The solution of differential equation :

$$t \cdot \frac{1}{x^6} = \int -3x^2 \cdot \frac{1}{x^6} dx$$

$$\Rightarrow \frac{1}{x^6 y} = \frac{-3}{-3x^3} + C$$

Putting $x = 1, y = \frac{1}{3}$

$$\Rightarrow 3 = 1 + C \quad \therefore C = 2$$

$$\Rightarrow \frac{1}{x^6 y} = \frac{1}{x^3} + 2$$

$$\Rightarrow y = \frac{\left(\frac{1}{x^6}\right)}{\frac{1}{x^3} + 2} = \frac{1}{x^3(1 + 2x^3)}$$

$$= \frac{1}{x^m(1 + n \cdot x^p)} \Rightarrow 3 = m, n = 2, p = 3$$

$$\therefore m^2 + n^2 + p^2 = 22$$

6. Ans. 3.00

$$\begin{aligned} \text{Sol. } (\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) &= \{(\vec{a} \times \vec{b}) \cdot \vec{d}\} \vec{c} - \{(\vec{a} \times \vec{b}) \cdot \vec{c}\} \vec{d} \\ &= m\vec{c} + n\vec{d} \end{aligned}$$

$$\therefore m + n = (\vec{a} \times \vec{b}) \cdot \vec{d} - (\vec{a} \times \vec{b}) \cdot \vec{c}$$

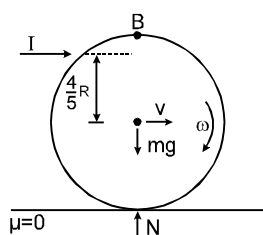
$$= (\vec{a} \times \vec{b}) \cdot (\vec{d} - \vec{c}) = -(\vec{a} \times \vec{b}) \cdot (\vec{c} - \vec{d})$$

$$= 3$$

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE-II)**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (A,B,D)

Sol. Using linear impulse momentum equation



$$I = mv \Rightarrow v = \frac{I}{M}$$

Using angular impulse momentum equation wrt centre

$$I \cdot \frac{4}{5} R = \frac{2}{5} MR^2 \omega$$

$$\omega = \frac{2I}{MR} \quad \dots\dots\dots(2)$$

Point B has to traverse angle π to reach ground.

$$\text{As } \omega \text{ is const., time required } t = \frac{\theta}{\omega} = \frac{\pi}{2I}$$

$$MR = \frac{MR\pi}{2I}$$

2. Ans. (A,C,D)

Sol. At max. amp. $v = 0$; $E = Ax^2 \Rightarrow x = \sqrt{\frac{E}{A}}$

$$\text{For max. vel. } x = 0 ; E = Bv^2 \Rightarrow v = \sqrt{\frac{E}{B}}$$

$$U = Ax^2$$

$$F = -\frac{\partial E}{\partial x} = -2Ax \Rightarrow F \text{ (at } x = 0) = 0$$

 $\therefore x = 0$ is eq. position.

$$F = -2Ax \Rightarrow a = \frac{-2Ax}{m}$$

$$\frac{1}{2}mv^2 = Bv^2 \Rightarrow m = 2B$$

$$\therefore a = -\frac{A}{B}x$$

3. Ans. (B,D)

Sol. It is better to make a chart for all the four cases which show the phase difference due to reflection at top and bottom surfaces of film.

	Top surfaces	Bottom surfaces	Due to $2L$	Total
Case I	0	0	6π	6π
Case II	0	π	6π	7π
Case III	0	π	6π	7π
Case IV	π	π	6π	8π

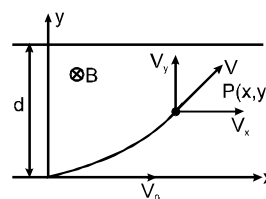
4. Ans. (A,D)

Sol. Let at time t particle be at point $P(x, y)$ and its velocity be

$$\vec{V} = (V_x \hat{i} + V_y \hat{j})$$

$$|\vec{V}| = |\vec{V}_0| \Rightarrow V_0^2 = V_x^2 + V_y^2$$

(work done by magnetic field is always zero so change in magnitude of velocity is zero)

Then, magnetic force on the particle at point P is

$$\vec{F} = q (V_x \hat{i} + V_y \hat{j}) \times B_0 \left(1 + \frac{y}{d}\right) (-\hat{k})$$

$$\Rightarrow -qB_0 \left[1 + \frac{y}{d}\right] dy = mdv_x$$

Now when the particle will be coming out of the at that point $y = d$. Let the velocity in x -direction be V_x then integrating we get,

$$\int_{v_0}^{v_x} dv_x = -\frac{qB_0}{m} \int_0^d \left[1 + \frac{y}{d}\right] dy$$

$$= -\frac{qB_0}{m} \left[d + \frac{d^2}{2d} \right] = -\frac{3qB_0 d}{2m}$$

$$\text{so } V_x = V_0 - \frac{3qB_0 d}{2m}$$

Now

$$\Rightarrow V_y = \sqrt{V_0^2 - V_x^2}$$

$$\Rightarrow V_y = \sqrt{V_0^2 - \left(V_0 - \frac{3qB_0 d}{2m} \right)^2}$$

5. Ans. (A,C)

Sol. Charge on capacitor before insertion of dielectric slab = 100 mC

Charge on capacitor after insertion of dielectric slab = 300 mC

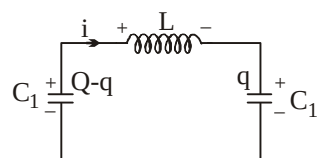
Increase in charge on the capacitor
= 300 - 100 = 200 mC

Heat produced = 0

Energy supplied by the cell = increase in stored potential energy + work done on the person who filling the dielectric slab.

6. Ans. (A,D)

Sol. at any time 't'



using KVL :

$$\frac{Q-q}{C_1} - L \frac{di}{dt} - \frac{q}{C_1} = 0$$

$$\text{or } \frac{Q-q}{C_1} - L \left(\frac{d^2 q}{dt^2} \right) - \frac{q}{C_1} = 0$$

$$\text{or } L \frac{d^2 q}{dt^2} = \frac{Q}{C_1} - \frac{2q}{C_1}$$

$$\text{or } \frac{d^2 q}{dt^2} = - \left(\frac{2}{LC_1} \right) q + \frac{Q}{LC_1}$$

$$\Rightarrow q = \frac{Q}{2} \sin(\omega t + \theta) + \frac{Q}{2} \left\{ \omega = \sqrt{\frac{2}{LC_1}} \right\}$$

$$Q_1 = Q - q = \frac{Q}{2} (1 + \cos \omega t)$$

$$= \frac{CV_0}{2} (1 + \cos \omega t)$$

$$Q_2 = q = \frac{Q}{2} (1 - \cos \omega t)$$

$$= \frac{CV_0}{2} (1 - \cos \omega t)$$

SECTION-II

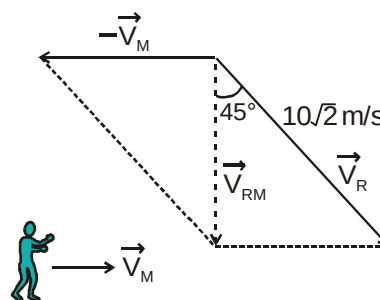
1. Ans. 15.00

Sol. In the first case :

From the figure it is clear that

$\vec{V}_{RM}$ is 10 m/s downwards and

$\vec{V}_M$ is 10 m/s towards right.



In the second case :

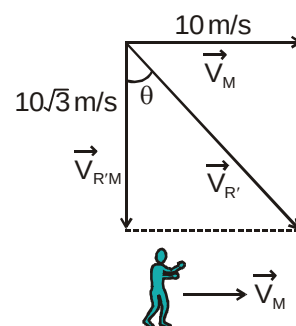
Velocity of rain as observed by man becomes $\sqrt{3}$ times in magnitude.

$\therefore$ New velocity of rain

$$\vec{V}_{R'} = \vec{V}_{R'M} + \vec{V}_M$$

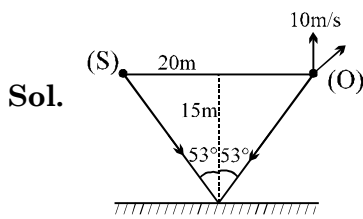
$\therefore$ The angle rain makes with vertical is

$$\tan \theta = \frac{10}{10\sqrt{3}} \quad \text{or } \theta = 30^\circ$$



$\therefore$ Change in angle of rain = $45 - 30 = 15^\circ$.

2. Ans. 8.00



Sol.

$$f_{\text{direct}} = 440 \text{ Hz}$$

$$f_{\text{indirect}} = 440 \left[\frac{330 - 10 \cos 53^\circ}{330} \right]$$

$$= 440 \left[1 - \frac{40}{3} \left(\frac{3}{5} \right) \right]$$

$$= 440 - 8 \Rightarrow 432 \text{ Hz}$$

$$\Delta f = 440 - 432 = 8 \text{ Hz}$$

3. Ans. 0.00

Sol. Let x is the vertical distance covered by

the ring. Then $x = L \tan 37^\circ = 0.7 \times \frac{3}{4}$

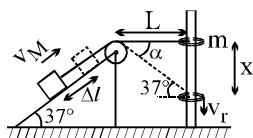
$$\Delta l = L \sec 37^\circ - L = L (\sec 37^\circ - 1)$$

$$\Rightarrow \frac{L}{4} = D l$$

Δl = distance moved by block M

Now, from constraint relation

$$v_M = v_r \sin 37^\circ = \frac{3}{5} v_r \quad \dots(1)$$



v_r = velocity of ring, v_M = velocity of the block at this instant

From mechanical energy conservation

$$\Delta PE + \Delta KE = 0$$

$$-mgx + Mg \Delta l \sin 37^\circ + \frac{1}{2} m v_r^2 + \frac{1}{2} M v_M^2 = 0 \quad \dots(2)$$

on solving eq. (1) and (2), we get

$$v_r = 0 \text{ m/s}$$

Hence instantaneous power will also be zero.

4. Ans. 7.66 to 7.67

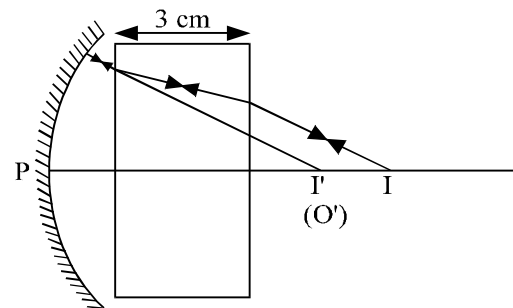
Sol. The rays starting from the object O would appear to come from the point O' , due to refraction through the glass plate.

This displacement OO' is given by $OO' = t (1 - 1/\mu)$ where t is the thickness of the plane and μ its refractive index.

$$\therefore OO' = 3 \left(1 - \frac{1}{1.5} \right) = 1 \text{ cm}$$

$$\therefore PO' = 21 - 1 = 20 \text{ cm}$$

Now if the object were at O' (apparent position) its image formed by the mirror would have been at a distance v from the mirror (at I') so that



$$v = \frac{uf}{u-f}, \quad u = 20 \text{ cm},$$

$$f = \frac{R}{2} = \frac{10}{2} = 5 \text{ cm}$$

$$v = \frac{20 \times 5}{20 - 5}$$

$$v = \frac{100}{15} = \frac{20}{3} \text{ cm}$$

However, the reflected rays have to pass through the glass slab and would converge at I .

$$\text{But } II' = 3 \left(1 - \frac{1}{1.5} \right) = 1 \text{ cm}$$

$\therefore$ The final position of the image will be at a distance $PI = \frac{20}{3} + 1 = 7.67 \text{ cm}$ from the mirror.

5. Ans. 35.00

$$\text{Sol. Total resistance} = \int_0^L \lambda x dx = \frac{\lambda}{2} L^2$$

For balance condition

$$\frac{R_1}{R_2} = \frac{(\lambda/2)l_1^2}{(\lambda/2)(L^2 - l_1^2)}$$

$$\text{Similarly, } \frac{R_2}{R_1} = \frac{l_2^2}{L^2 - l_2^2}$$

$$\therefore \frac{R_1}{R_2} = \frac{l_1^2}{L^2 - l_1^2} = \frac{l_2^2}{L^2 - l_2^2} = n$$

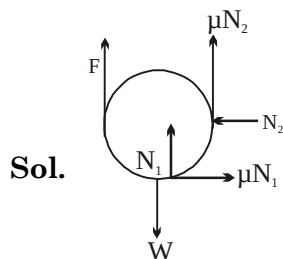
$$\text{Solving, } l_1 = L\sqrt{\frac{n}{n+1}} \text{ and } l_2 = L\sqrt{\frac{1}{n+1}}$$

$$\therefore \Delta = l_1 - l_2 = \frac{L}{\sqrt{n+1}}(\sqrt{n} - 1)$$

Putting $L = 100$ cm and $n = 3$

$$D = \frac{100}{2}(\sqrt{3} - 1) = 35 \text{ cm}$$

6. Ans. 8.00



Sol.

$$\Sigma F_y = 0 \Rightarrow F + \mu N_2 + N_1 = W$$

$$\Sigma F_x = 0 \Rightarrow F = \mu N_1$$

$$\Sigma \tau = 0 \Rightarrow Fr = \mu N_1 r + \mu N_2 r$$

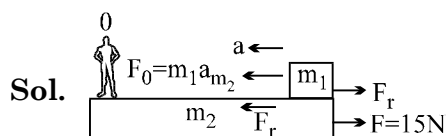
Solving above equations with $\mu = 0.5$

$$\Rightarrow F = \frac{3W}{8} = 0.375W$$

after rounding-off upto 2 decimal places, it will be 0.38W.

SECTION-III

1. Ans. 5



Sol.

$$Fr = mmg = 0.1 \times 1 \times 10 = 1\text{N}$$

$$a_{m_2} = \frac{F - Fr}{m_2} = \frac{15 - 1}{10} = 1.4 \text{ m/sec}^2$$

$$a_{m_1, m_2} = a = \frac{m_1 a_{m_2} - Fr}{m_1} = \frac{1.4 - 1}{1} = 0.4 \text{ m/sec}^2$$

Time taken by m_1 to move 1 m

$$t = \sqrt{\frac{2 \times 1}{0.4}} = \sqrt{5}$$

2. Ans. 2

Sol. Using the energies for the photons for two given transitions,

$$E_{2n} - E_n = 40.8 \Rightarrow 13.6z^2 \left[\frac{1}{n^2} - \frac{1}{4n^2} \right] = 40.8$$

and

$$E_{2n} - E_1 = 204 \Rightarrow 13.6z^2 \left[\frac{1}{1^2} - \frac{1}{4n^2} \right] = 204$$

3. Ans. 8

$$\text{Sol. } \frac{PT^2}{V} = k \Rightarrow \frac{nRT}{V} \cdot \frac{T^2}{V} = k \Rightarrow T^3 = \frac{kV^2}{nR}$$

Now, Differentiating,

$$3T^2 \frac{dT}{dV} = \frac{2kV}{nR} \Rightarrow \frac{dV}{dT} = \frac{3nRT^2}{2kV} = \frac{3}{2} \frac{nR}{P}$$

Molar Heat capacity,

$$C = C_V + \frac{P}{n} \frac{dV}{dT} = \frac{5R}{2} + \frac{P}{n} \cdot \frac{3}{2} \frac{nR}{P} = \frac{8R}{2}$$

4. Ans. 4

Sol. For constant velocity,

$$a = 0$$

$$F_0 = F_m$$

$$= i\ell B = \left(\frac{\varepsilon}{R} \right) \ell B = \left(\frac{B\ell v_0}{R} \right) \ell B$$

$$v_0 = \frac{F_0 R}{B^2 \ell^2} \text{ velocity at point 'P'}$$

$$\text{Now, retardation } a = \frac{F_m}{m} = \frac{i\ell B}{m}$$

$$a = \frac{B^2 \ell^2}{mR} v$$

$$\Rightarrow -v \frac{dv}{ds} = \frac{B^2 \ell^2}{mR} v$$

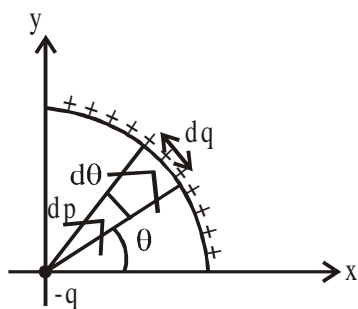
$$\text{or } -\int_{v_0}^0 dv = \frac{B^2 \ell^2}{mR} \int_0^s ds$$

$$\text{or } v_0 = \frac{B^2 \ell^2}{mR} s$$

$$\text{or } s = \frac{mRv_0}{B^2 \ell^2} = \frac{F_0 m R^2}{B^4 \ell^4} = 320 \text{ m}$$

5. Ans. 8

Sol. Dipole moment $\frac{2\sqrt{2}qR}{\pi}$



$$|dp| = R dq = R \times \frac{q}{\frac{\pi}{2}R} (R d\theta) = \frac{2q}{\pi} R d\theta$$

$$p_x = \int_0^{\pi/2} dp \cos \theta = \frac{2q}{\pi} R \int_0^{\pi/2} \cos \theta d\theta = \frac{2qR}{\pi} \hat{i}$$

$$\vec{p}_y = \int_0^{\pi/2} dp \sin \theta = \frac{2q}{\pi} R \int_0^{\pi/2} \sin \theta d\theta = \frac{2qR}{\pi} \hat{j}$$

$$|p| = \sqrt{p_x^2 + p_y^2} = \sqrt{2} \times \frac{2qR}{\pi} = \frac{2\sqrt{2} \times 1 \times \sqrt{8}\pi}{\pi} = 8\text{cm}$$

6. Ans. 5

Sol. By Energy Conservation

$$\frac{mgR}{\sqrt{2}} = \frac{1}{2} m (\sqrt{2}R)^2 \omega^2$$

$$\Rightarrow \omega^2 = \frac{3g}{\sqrt{2}R}$$

$$\text{Now, } 2N \cos 45^\circ - mg = m \times \frac{3g}{\sqrt{2}R} \times \frac{R}{\sqrt{2}}$$

$$\Rightarrow N = \frac{5mg}{2\sqrt{2}} = 50$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (A,C,D)

2. Ans. (A,B,C)

3. Ans. (A,B,D)

4. Ans. (A,B,C,D)

(i) $\text{H}_2\text{A} + \text{NaOH} \longrightarrow \text{NaHA} + \text{H}_2\text{O}$

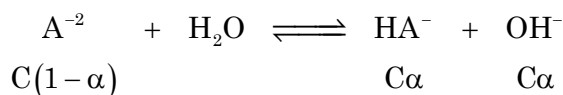
0.1M, 1mole 0.1M, 1mole

— — 0.05M, 200ml

$$\text{PH} = \frac{\text{PKa}_1 + \text{PKa}_2}{2} = \frac{3+6}{2} = 4.5$$

(ii) $\text{H}_2\text{A} + 2\text{NaOH} \longrightarrow \text{Na}_2\text{A} + \text{H}_2\text{O}$

0.1 M, 100 ml 0.1 M, 200ml $M = \frac{10}{300} = \frac{1}{30}$



$$K_h = \frac{[\text{HA}^-][\text{OH}^-][\text{H}^+]}{[\text{A}^{2-}][\text{H}^+]} = \frac{K_w}{K_{a_2}} = \frac{10^{-14}}{10^{-6}} = 10^{-8}$$

$$K_h = \frac{C\alpha^2}{(1-\alpha)} = 10^{-8}$$

$$\Rightarrow \frac{10}{300} \alpha^2 = 10^{-8}$$

$$\alpha^2 = 30 \times 10^{-8}$$

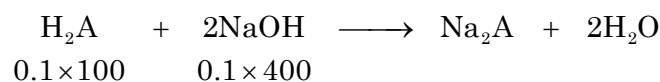
$$\alpha^2 = \sqrt{30} \times 10^{-4}$$

$$[\text{OH}^-] = \alpha = \frac{1}{30} \cdot \sqrt{30} \times 10^{-4} = \frac{1}{\sqrt{30}} \times 10^{-4}$$

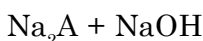
$$[\text{OH}^+] = \frac{10^{-14}}{10^{-4} \times \frac{1}{\sqrt{30}}} = \sqrt{30} \times 10^{-10}$$

$$\text{PH} = 10 - \frac{1}{2} \log 30 = 10.738 = 9.26$$

(iii)



10 40
0 20 20



$$(\text{OH}^-) = \frac{1}{25}$$

$$\text{POH} = -\log \frac{1}{25} = \log 25$$

$$\text{POH} = 1.3979$$

$$\text{PH} = 12.602$$

(iv) Buffer solution

5. Ans. (A,B,C,D)

6. Ans. (B,C)

SECTION-II

1. Ans. (5.00)
2. Ans. (0.00)
3. Ans. (0.00)
4. Ans. (0.02)

$$Z = 1 + \left(b - \frac{a}{RT} \right) \frac{1}{V_m}$$

$$PV_m = ZRT \Rightarrow \frac{1}{V_m} = \frac{P}{ZRT}$$

$$Z = 1 + \frac{P}{ZRT} \left(b - \frac{a}{RT} \right) \left(T_B = \frac{a}{Rb} \right)$$

$$Z = 1 + \frac{P}{ZRT} \left(b - \frac{bT_B}{T} \right)$$

$$\frac{Z(Z-1)RT}{P} = b \left(1 - \frac{T_B}{T} \right) = b \left(\frac{T - T_B}{T} \right)$$

$$b = \left(\frac{T}{T - T_B} \right) \cdot \frac{(Z-1)ZRT}{P}$$

$$b = \left(\frac{300}{300 - 200} \right) \cdot \frac{0.0003 \times 1.0003}{1 \text{ bar}} \times \frac{0.08 \text{ litre bar x}^{-1} \text{ mole}^{-1}}{300}$$

$$b = 3 \times 3 \times 10^{-4} \times 1.003 \times 8 \times 10^{-2} \times 300$$

$$b = 0.02160 \text{ litre/mole}$$

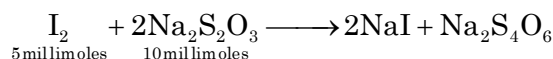
$$b \approx 0.02 \text{ litre / mole}$$

5. Ans. (2.00)

6. Ans. (6.00)

SECTION-III

1. Ans. (5)



$$x = \frac{25}{3} \text{ and } y = 10$$

$$3X - 2Y = 5$$

2. Ans. (2)

3. Ans. (3)

4. Ans. (9)

5. Ans. (8)

6. Ans. (6)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (B,D)

Sol. There will be three points in a plane at a time and a plane parallel to the plane containing three points and equidistant from point and plane. No. of such plane '4'. Now consider set of skew lines passing through two points at a time. Three sets of two skew lines can be obtained. Now consider plane bisecting line of shortest distance between them, such three planes are possible. Total planes '7'.

2. Ans. (A,C)

Sol. Divide throughout by x^3y , we get

$$2 \frac{dx}{x} + 2 \frac{dy}{y} - 2 \frac{y^3}{x^3} dx + 3 \frac{y^2}{x^2} dy = 0$$

$$\Rightarrow 2 \frac{dx}{x} + 2 \frac{dy}{y} + \frac{3xy^2dy - 2y^3dx}{x^3} = 0$$

$$\Rightarrow 2 \frac{dx}{x} + 2 \frac{dy}{y} + \frac{3x^2y^2dy - 2xy^3dx}{x^4} = 0, \text{ on integrating we get}$$

$$\Rightarrow 2(\ln x + \ln y) + \frac{y^3}{x^2} = C \text{ Now put } x = 1, y = 1$$

$$\Rightarrow 2\ln(1)+1=C \Rightarrow C=1 \Rightarrow 2\ln(xy)+\frac{y^3}{x^2}=1, \quad \begin{matrix} m=3 \\ n=2 \end{matrix}$$

3. Ans. (A,B,C)

Sol. We have $\int_0^x f(t)dt + x \int_0^x f(t)dt - \int_0^x t \cdot f(t)dt = (-1 + e^{-x})$

Differentiate both sides with respect to x , we get

$$f(x) + \int_0^x f(t)dt = -e^{-x} \quad \dots(i)$$

Put $x=0$, we get $f(0) = -1$

Again differentiating both sides of equation (i) with respect to ' x ', we get

$$e^x (f(x) + f'(x)) = 1$$

Now integrating both sides with respect to x , we get

$$e^x f(x) = x + C, f(0) = -1$$

$$-1 = 0 + C \Rightarrow C = -1$$

$$f(x) = (x-1)e^{-x}, \text{ Now we can check options.}$$

4. Ans. (A,B,D)

Sol. $\phi(x) = f(x) + Ag(x)$

clearly $\phi(x)$ is continuous and differentiable in $[x_1, x_2]$ and (x_1, x_2) respectively.

$$\text{If } f(x_1) + Ag(x_1) = f(x_2) + Ag(x_2)$$

$$\Rightarrow \frac{f(x_1) - f(x_2)}{g(x_2) - g(x_1)} = A$$

$$\Rightarrow A = \frac{4-6}{2-1} = \frac{-2}{1} = -2$$

We can apply Rolle's theorem on $\phi(x) = f(x) - 2g(x)$

which states that $\phi'(c) = 0$ for some $c \in (x_1, x_2)$

$$f'(c) = 2g'(c) \Rightarrow \lambda = 2$$

(A) Clearly prime number has two divisors

(B) No. of ways of putting 5 identical balls in three identical boxes (3,1,1) & (2,2,1) two ways.

$$(C) e^{\sin x} - x = 0 \Rightarrow \sin x = \ln x$$

only '1' solution so this option is incorrect.

(D) No. of into functions are two.

5. Ans. (C,D)

Sol. $f(0) = \lim_{h \rightarrow 0^+} f(0+h) = \lim_{h \rightarrow 0^-} f(0-h)$

$$a = \ln(1) + b(0) \Rightarrow \boxed{a=0}$$

$$f'(x) = \begin{cases} (\cos(\sin x)) \cos x & ; x \geq 0 \\ -\frac{\sin x}{\cos x} + b & ; x < 0 \end{cases}$$

$$\text{LHD} = (\cos(0)) \cos 0 = 1$$

$$\text{RHD} = a + b = b$$

$$\Rightarrow b = 1$$

6. Ans. (B,C,D)

$$\text{Sol. } T_r \text{ (r}^{\text{th}} \text{ term of the series)} = (-1)^{r-1} \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{r} \right) \cdot {}^nC_r$$

$$\text{Now } (-1)^{r-1} (1 + x + x^2 + \dots + x^{r-1}) \cdot {}^nC_r = \left(\frac{1-x^r}{1-x} \right) (-1)^{r-1} \cdot {}^nC_r$$

$$\text{Now } \frac{1}{(1-x)} \sum_{r=1}^n \left[(-1)^{r-1} \cdot {}^nC_r - (-1)^{r-1} \cdot x^r \cdot {}^nC_r \right]$$

$$= \frac{1}{(1-x)} \left[(-1) \sum_{r=1}^n (-1)^r \cdot {}^nC_r + \sum_{r=1}^n (-1)^r \cdot x^r \cdot {}^nC_r \right]$$

$$= \frac{1}{(1-x)} \left[(-1) \left\{ (1-1)^n - 1 \right\} + \left\{ (1-x)^n - 1 \right\} \right]$$

$$= \frac{(1-x)^n}{(1-x)} = (1-x)^{n-1}$$

$$\Rightarrow \sum_{r=1}^n (-1)^{r-1} (1 + x + x^2 + \dots + x^{r-1}) \cdot {}^nC_r = (1-x)^{n-1}$$

Integrating above equation, we get

$$\int_0^1 \left[\sum_{r=1}^n (-1)^{r-1} (1 + x + x^2 + \dots + x^{r-1}) \cdot {}^nC_r \right] dx = \int_0^1 (1-x)^{n-1} dx$$

$$\Rightarrow \sum_{r=1}^n (-1)^{r-1} \cdot {}^nC_r \int_0^1 (1 + x + x^2 + \dots + x^{r-1}) dx = \left[-\frac{(1-x)^n}{n} \right]_0^1$$

$$\Rightarrow \sum_{r=1}^n (-1)^{r-1} \cdot {}^nC_r \left[x + \frac{x^2}{2} + \frac{x^3}{3} + \dots + \frac{x^r}{r} \right]_0^1 = \frac{1}{n}$$

$$\Rightarrow \sum_{r=1}^n (-1)^{r-1} \cdot {}^nC_r \left[1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{r} \right] = \frac{1}{n}$$

SECTION-II

1. Ans. 0.45

Sol. Applying $R_1 \rightarrow R_1 - R_2 \sec x$ and $R_2 \rightarrow R_2 - R_3 \cos^2 x$

$$f(x) = \begin{vmatrix} 0 & 0 & \sec^2 x + \frac{\cos x}{\sin^2 x} - \frac{1}{\cos x \sin^2 x} \\ 0 & \cos^2 - \cos^4 x & \operatorname{cosec}^2 x - \cos^4 x \\ 1 & \cos^2 x & \cos^2 x \end{vmatrix}$$

Expanding along C_1

$$f(x) = -(\cos^2 x - \cos^4 x) \left(\sec^2 x + \frac{\cos x}{\sin^2 x} - \frac{1}{\cos x \cdot \sin^2 x} \right)$$

$$= -\cos^2 x \cdot \sin^2 x \left(\frac{1}{\cos^2 x} + \frac{\cos x}{\sin^2 x} - \frac{1}{\cos x \sin^2 x} \right)$$

$$= -\cos^2 x \cdot \sin^2 x \left(\frac{\sin^2 x + \cos^3 x - \cos x}{\cos^2 x \sin^2 x} \right)$$

$$= -(\sin^2 x + \cos x (\cos^2 x - 1))$$

$$= -(\sin^2 x - \cos x \cdot \sin^2 x)$$

$$= \cos x \cdot \sin^2 x - \sin^2 x$$

$$= \int_0^{\frac{\pi}{2}} (\cos x \cdot \sin^2 x - \sin^2 x) dx$$

$$= \int_0^{\frac{\pi}{2}} \cos x \cdot \sin^2 x dx - \int_0^{\frac{\pi}{2}} \sin^2 x \cdot dx$$

$$\sin x = t$$

$$\cos x dx = dt$$

$$\int_0^1 t^2 \cdot dt - \frac{1}{2} \cdot \frac{\pi}{2}$$

$$= \frac{1}{3} - \frac{\pi}{4}$$

$$= \frac{4-3\pi}{12} = -0.45$$

$$|-0.45| = 0.45$$

2. **Ans. 45.95**

Sol. $x^4 - 4x^3 + 6x^2 - 4x + 1 = 2021$

$$(x-1)^4 = 2021$$

$$(x-1)^2 = \pm\sqrt{2021}$$

$$(x-1)^2 = -\sqrt{2021} \text{ because only non-real roots required}$$

$$x^2 + 1 - 2x + \sqrt{2021} = 0$$

$$x^2 - 2x + 1 + \sqrt{2021} = 0, P = \text{product of roots} = 1 + \sqrt{2021} = 45.95$$

3. **Ans. 31.25**

Sol. $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c} \leq 1 \Rightarrow \frac{1}{1+b} + \frac{1}{1+c} \leq 1 - \frac{1}{1+a} = \frac{a}{1+a}$

$$\frac{a}{1+a} \geq \frac{1}{1+b} + \frac{1}{1+c}$$

$$\text{Similarly } \frac{b}{1+b} \geq \frac{1}{1+c} + \frac{1}{1+a} \text{ and } \frac{c}{1+c} \geq \frac{1}{1+a} + \frac{1}{1+b}$$

Apply A.M ≥ G.M. for $\frac{1}{1+b} + \frac{1}{1+c}$

$$\therefore \frac{1}{1+b} + \frac{1}{1+c} \geq \frac{2}{\sqrt{1+b}\sqrt{1+c}} \Rightarrow \frac{a}{1+a} \geq \frac{2}{\sqrt{(1+b)(1+c)}}$$

$$\text{similarly } \frac{b}{1+b} \geq \frac{2}{\sqrt{(1+c)(1+a)}} \text{ and } \frac{c}{1+c} \geq \frac{2}{\sqrt{(1+a)(1+b)}}$$

Multiply the results

$$\left(\frac{a}{1+a}\right)\left(\frac{b}{1+b}\right)\left(\frac{c}{1+c}\right) \geq \left(\frac{2}{\sqrt{(1+b)(1+c)}}\right)\left(\frac{2}{\sqrt{1+c}\sqrt{1+a}}\right)\left(\frac{2}{\sqrt{1+a}\sqrt{1+b}}\right)$$

$$\Rightarrow abc \geq 8$$

$$\begin{aligned} \text{Now expand } (1+a^2)(1+b^2)(1+c^2) \\ = 1 + (a^2 + b^2 + c^2) + (a^2b^2 + b^2c^2 + c^2a^2) + a^2b^2c^2 \\ \geq 1 + 3(a^2b^2c^2)^{\frac{1}{3}} + 3(a^4b^4c^4)^{\frac{1}{3}} + a^2b^2c^2 \\ \geq 1 + 3(2^2) + 3(2^4) + (8)^2 \\ \geq 1 + 12 + 48 + 64 \end{aligned}$$

$$\geq 125 = \lambda \quad \text{so } \frac{\lambda}{4} = 31.25$$

4. Ans. 1.25

Sol. $z_1^3 - 3z_1z_2^2 = 2 \quad \dots(1)$

$$3z_1^2z_2 - z_2^3 = 11 \quad \dots(2)$$

Multiply equation (2) by i and add in equation (1)

$$z_1^3 - 3z_1z_2^2 + 3iz_2z_1^2 - iz_2^3 = 2 + 11i$$

$$z_1^3 + i^2 3z_1z_2^2 + 3iz_2z_1^2 + (iz_2)^3 = 2 + 11i$$

$$(z_1 + iz_2)^3 = 2 + 11i$$

$$\text{similarly } (z_1 - iz_2)^3 = 2 - 11i$$

$$\text{Multiply the two } ((z_1 + iz_2)(z_1 - iz_2))^3 = 4 + 121$$

$$(z_1^2 + z_2^2)^3 = 125$$

$$z_1^2 + z_2^2 = 5$$

$$|z_1^2 + z_2^2| = 5$$

$$\frac{|z_1^2 + z_2^2|}{4} = \frac{5}{4} = 1.25$$

5. Ans. 1.00

Sol. A tangent to $\frac{x^2}{16} - \frac{y^2}{9} = 1$ is

$$y = m_1x \pm \sqrt{16m_1^2 - 9} \quad \dots\dots(i)$$

The other hyperbola is $\frac{x^2}{-9} - \frac{y^2}{-16} = 1$

then tangent to it is

$$y = m_2 x \pm \sqrt{-9m_2^2 - (-16)} \quad \dots\dots(ii)$$

(i) & (ii) are same.

$$\Rightarrow m_1 = m_2 \quad \& \quad 16m_1^2 - 9 = -9m_2^2 + 16$$

$$\Rightarrow 25m_1^2 = 25 \quad \Rightarrow \quad m_1^2 = 1 \quad \Rightarrow \quad m_1 = \pm 1$$

$$\therefore m = 1$$

6. Ans. 1.75

Sol. Let a circle be $x^2 + y^2 + 2gx + 2fy + c = 0$, which intersects the ellipse at $(a \cos \phi, b \sin \phi)$, then

$$a^2 \cos^2 \phi + b^2 \sin^2 \phi + 2ga \cos \phi + 2fb \sin \phi + c = 0$$

$$a^2 \left(\frac{1 - \tan^2 \frac{\phi}{2}}{1 + \tan^2 \frac{\phi}{2}} \right)^2 + b^2 \left(\frac{2 \tan \frac{\phi}{2}}{1 + \tan^2 \frac{\phi}{2}} \right)^2 + 2ga \left(\frac{1 - \tan^2 \frac{\phi}{2}}{1 + \tan^2 \frac{\phi}{2}} \right) + 2fb \left(\frac{2 \tan \frac{\phi}{2}}{1 + \tan^2 \frac{\phi}{2}} \right) + c = 0$$

$$\Rightarrow a^2 \left(1 - 2 \tan^2 \frac{\phi}{2} + \tan^4 \frac{\phi}{2} \right) + 4b^2 \tan^2 \frac{\phi}{2} + 2ga \left(1 - \tan^2 \frac{\phi}{2} \right) + 2fb \left(2 \tan \frac{\phi}{2} + 2 \tan^3 \frac{\phi}{2} \right)$$

$$+ c \left(1 + 2 \tan^2 \frac{\phi}{2} + \tan^4 \frac{\phi}{2} \right) = 0$$

$$\Rightarrow (a^2 - 2ga + c) \tan^4 \frac{\phi}{2} + 4fb \tan^3 \frac{\phi}{2} + (4b^2 - 2a^2 + 2c) \tan^2 \frac{\phi}{2} + 4fb \tan \frac{\phi}{2} + (a^2 + 2ga + c) = 0 \quad \dots(1)$$

clearly $\phi_1, \phi_2, \phi_3, \phi_4$ are the four values of ϕ obtained from (1)

$$\text{So, } \sum \tan \frac{\phi_1}{2} = -\frac{4fb}{a^2 - 2ga + c}, \sum \tan \frac{\phi_1}{2} \tan \frac{\phi_2}{2} = \frac{4b^2 - 2a^2 + 2c}{a^2 - 2ga + c}$$

$$\sum \tan \frac{\phi_1}{2} \cdot \tan \frac{\phi_2}{2} \cdot \tan \frac{\phi_3}{2} = -\frac{4fb}{a^2 - 2ga + c}$$

$$\text{and } \tan \frac{\phi_1}{2} \cdot \tan \frac{\phi_2}{2} \cdot \tan \frac{\phi_3}{2} \cdot \tan \frac{\phi_4}{2} = \frac{a^2 + 2ga + c}{a^2 - 2ga + c}$$

$$\tan \left(\frac{\phi_1}{2} + \frac{\phi_2}{2} + \frac{\phi_3}{2} + \frac{\phi_4}{2} \right) = \frac{s_1 - s_3}{1 - s_2 + s_4} = 0 \quad \text{since } s_1 = s_3$$

$$\frac{\phi_1}{2} + \frac{\phi_2}{2} + \frac{\phi_3}{2} + \frac{\phi_4}{2} = n\pi \quad \therefore \phi_1 + \phi_2 + \phi_3 + \phi_4 = 2n\pi$$

$$\cos(\phi_1 + \phi_2 + \phi_3 + \phi_4) = 1$$

SECTION-III

1. Ans. 0

Sol. Equation of normal at $P(3 \cos \theta, 2 \sin \theta)$ is

$$3x \sec \theta - 2y \operatorname{cosec} \theta = 5$$

Now it is tangent to circle so

$$\frac{5}{\sqrt{9\sec^2\theta + 4\operatorname{cosec}^2\theta}} = \sqrt{3}$$

$$\text{But } \{9\sec^2\theta + 4\operatorname{cosec}^2\theta\} = (3+2)^2 = 25$$

$\therefore$ No such θ exist.

2. **Ans. 1**

Sol. Let θ be the angle between OA and OB, then

$$\cos\theta = \frac{OA^2 + OB^2 - AB^2}{2 \cdot OA \cdot OB}$$

Since OA, AB and OB are in A.P.

$$2AB = OA + OB$$

$$\Rightarrow 3AB^2 - 2OA \cdot OB = OA^2 + OB^2 - AB^2$$

$$\text{From (i) and (ii) } \cos\theta = \frac{3AB^2}{2OA \cdot OB} - 1$$

$$\text{Using } A \cdot M \geq G \cdot M; \quad \frac{AB^2}{OA \cdot OB} \geq 1$$

$$\therefore \cos\theta \geq \frac{1}{2} \Rightarrow \theta \leq 60^\circ$$

$$\tan\theta = \left| \frac{m-n}{1+mn} \right| \leq \sqrt{3} \Rightarrow |m-n| \leq \sqrt{3}|1+mn|, K = \sqrt{3}, [K] = 1$$

3. **Ans. 0**

$$\text{Sol. } \sin x \cdot \sin\left(\frac{1}{x}\right) = 1$$

$$\sin\left(\frac{1}{x}\right) = \operatorname{cosec} x$$

$$\text{clearly } \sin\left(\frac{1}{x}\right) = \operatorname{cosec} x = 1 \text{ or } \sin\left(\frac{1}{x}\right) = \operatorname{cosec} x = -1$$

which is not possible for same value of x .

4. **Ans. 3**

Sol. $x \in \text{prime and } x < 10$

$$\Rightarrow x = 2, 3, 5, 7$$

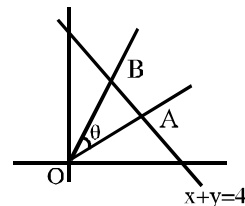
$$\text{Total no. of order pair } n(A) = 4 \times 10 = 40$$

$$\text{and } x^2 - 3y^2 = 1$$

$$\Rightarrow x^2 = 3y^2 + 1$$

For above condition to be satisfied only two such pairs (2,1) and (7,4) are possible

$$\Rightarrow P(A) = \frac{2}{40} = \frac{1}{20} = P \quad \text{So } 60P = 3$$



5. Ans. 7

$$AB^2 = BA \Rightarrow B^2 = A^{-1}BA \text{ squaring both the sides}$$

$$B^4 = (A^{-1}BA)(A^{-1}BA) = A^{-1}B^2A = A^{-1}(A^{-1}BA)A$$

$$B^4 = (A^{-1})^2 BA^2$$

$$\text{Again squaring } B^8 = ((A^{-1})^2 BA^2)((A^{-1})^2 BA^2)$$

$$\Rightarrow B^8 = (A^{-1})^2 B^2 A^2 = (A^{-1})^2 (A^{-1}BA) A^2 = (A^{-1})^3 BA^3$$

$$\text{Similarly } B^{64} = (A^{-1})^6 BA^6 \Rightarrow B^{63} = I$$

$$\text{so } m = 63 = k, \frac{k}{9} = 7$$

6. Ans. 2

$$\text{Let } \frac{I}{8} = \int_0^{\pi/2} \frac{x^2 (\cos x - \sin x) dx}{(\cos x + \sin x)^3}$$

$$\Rightarrow \frac{I}{8} = \int_0^{\pi/2} \frac{x^2 \cdot \cos 2x dx}{(1 + \sin 2x)^2}$$

$$\text{Let } 2x = t$$

$$\Rightarrow I = \int_0^{\pi} \frac{t^2 \cos t dt}{(1 + \sin t)^2}$$

By parts

$$= \left(\frac{-t^2}{1 + \sin t} \right)_0^{\pi} + 2 \int_0^{\pi} \frac{t}{1 + \sin t}$$

$$I = -\pi^2 + 2I_1$$

$$\text{where } I_1 = \int_0^{\pi} \frac{t}{1 + \sin t} \dots\dots(i)$$

From king property

$$I_1 = \int_0^{\pi} \frac{\pi - t}{1 + \sin t} \dots(ii)$$

(i) + (ii)

$$\Rightarrow 2I_1 = \pi \int_0^{\pi} \frac{dt}{1 + \sin t}$$

$$\Rightarrow 2I_1 = \pi \int_0^{\pi} (\sec^2 t - \sec t - \tan t) dt$$

$$\Rightarrow \frac{2I_1}{\pi} = (\tan t - \sec t)_0^{\pi} \Rightarrow \frac{2I_1}{\pi} = 1 + 1$$

$$\Rightarrow I_1 = \pi \Rightarrow I = \pi(2 - \pi)$$